

Challenge (Habanero)

- Q1.** Simplify $\sqrt{16x^2}$ where $x > 0$.
(A) $4x$
(B) $4\sqrt{x}$
(C) $\sqrt{4x}$
(D) $2\sqrt{2x}$
(E) $\sqrt{2x}$
- Q2.** Convert $\sqrt[3]{8x^3}$ to rational exponent form.
(A) $2x$
(B) $2x^{1/3}$
(C) $2x^3$
(D) $8^{1/3}x$
(E) $8x^{1/3}$
- Q3.** A recipe for making cookies calls for $\sqrt{50}$ cups of flour. Simplify this measurement.
(A) $5\sqrt{2}$
(B) $7\sqrt{2}$
(C) $25\sqrt{2}$
(D) $50\sqrt{2}$
(E) $10\sqrt{2}$
- Q4.** A building is designed with a height of $\sqrt[3]{216}$ meters. What is the height in simplest radical form?
(A) 6
(B) $36^{1/3}$
(C) $6^{1/3}$
(D) $\sqrt[3]{36}$
(E) $\sqrt[3]{6}$
- Q5.** A chef needs to convert a recipe that requires $\sqrt{24}$ tablespoons of an ingredient. Simplify this measurement.
(A) $\sqrt{4} \cdot \sqrt{6}$
(B) $\sqrt{6} \cdot \sqrt{4}$
(C) $2\sqrt{6}$
(D) $\sqrt{6} \cdot 2$
(E) $6\sqrt{2}$
- Q6.** A carpenter needs to cut a beam of length $\sqrt{32}$ meters. What is the length in simplest radical form?
(A) $4\sqrt{2}$
(B) $8\sqrt{2}$
(C) $2\sqrt{8}$
(D) $16\sqrt{2}$
(E) $\sqrt{16}$
- Q7.** Simplify $\sqrt{75x^2}$ where $x > 0$.
(A) $5x\sqrt{3}$
(B) $5x\sqrt{2}$
(C) $10x\sqrt{3}$
(D) $\sqrt{25x^2} \cdot \sqrt{3}$
(E) $\sqrt{3x^2} \cdot \sqrt{25}$
- Q8.** A bridge is designed with a width of $\sqrt[3]{125}$ meters. What is the width in simplest radical form?
(A) 5
(B) $5^{1/3}$
(C) $\sqrt[3]{5}$
(D) $25^{1/3}$
(E) $\sqrt[3]{25}$

Open Ended Questions (FRQ)

- Q9.** Simplify the expression $\sqrt{48x^2y}$, and then convert it to rational exponent form.
- Q10.** A construction plan requires a room with dimensions $\sqrt[3]{64}$ meters by $\sqrt{16}$ meters. Simplify these measurements and calculate the area of the room.

Chef's Tips

- Read each question carefully
- Use the space provided for calculations
- Answer all questions to the best of your ability